Ali Ezzat, PhD

Senior / Principal Data Scientist — Forecasting, Optimization, Recommender Systems & Applied Deep Learning — Retail, Clean Energy & Sustainability

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Cairo, Egypt — Open to relocation (Europe, Japan)

SUMMARY

- Senior/Principal Data Scientist and PhD with **7+ years** leading end-to-end analytics engagements — time-series **forecasting**, **optimization**, **recommender systems**, causal inference and generative AI — across energy-intensive industrial manufacturing, supply chain, consumer/retail-facing services and finance.
- Delivered measurable operational and sustainability impact: **23% lower industrial energy (natural gas) consumption** with zero quality failures, **18% lower supply-network cost**, cash-forecasting error cut **15x** (300%→20%), and customer-targeting precision lifted **3%→25%**.
- Core toolkit — demand forecasting, network/inventory optimization, dynamic pricing, recommender systems, and churn/lead-scoring — maps directly onto **retail merchandising, demand planning, and clean-energy operations analytics**.
- Deep expertise in deep learning (PyTorch, Transformers), LLM/GenAI system design, and end-to-end production ML deployment (Docker, APIs, MLOps).
- Founded and scaled a **10-person data science team**; owned hiring, technical direction, and mentoring across client-facing production ML projects.
- Open to relocation to **Europe or Japan**.

WORK EXPERIENCE

Principal Data Scientist *Mar 2019 – Present*

Synapse Analytics — Cairo, Egypt

- Founded and scaled the Data Science team from **1 to 10** data scientists; owned the hiring process end to end (resume screening, technical assessments, interviews).
- Led technical direction while remaining hands-on across production ML projects spanning consumer/retail analytics, energy-intensive manufacturing, supply chain, healthcare and finance.
- Designed production ML pipelines, Dockerized model APIs, and deployment workflows in partnership with data engineering teams.
- Presented proof-of-concept demos to prospective clients and ran controlled experiments (A/B testing) to validate model improvements before deployment.
- Served as applied ML lead for **Konan**, the company's internal MLOps platform, translating production workflows into platform requirements — Konan now supports **~25 production models** across **~15 client deployments**.
- Mentored junior data scientists through model development and deployment.
- *Technologies: Python, R, SQL, Docker, Git, PyTorch, Transformers, XGBoost, scikit-learn, ggplot2, Plotly, Shiny*

Industrial Energy Optimization — Unilever (Clean Energy / Sustainability)

- Designed and deployed a real-time ML recommendation system on live sensor data (10+ sensors) from a detergent-manufacturing tower, recommending nozzle, burner and slurry-feed settings each cycle.
- Cut monthly **natural-gas consumption by 23%** using nearest-neighbor retrieval over historical operating logs to surface past machine states with favorable energy profiles under matching conditions.
- Built automated quality safeguards that handed control to rule-based corrective actions whenever moisture content or bulk density drifted out of specification, resulting in **zero production batch failures**.
- Separately built a multi-variant manufacturing scheduler generating time-minimizing production schedules from a monthly demand sheet while respecting line-speed, efficiency and specification constraints.

Supply Network & Logistics Optimization — Procter & Gamble (Retail / Sustainability)

- Built an optimization platform recommending warehouse/hub locations, delivery routes, fleet requirements and shipment frequencies to minimize total logistics cost and distance traveled.
- Reduced **total supply-network cost by 18%** versus the client's existing network, using k-means clustering to assign stores to warehouses and a traveling-salesman algorithm to optimize milk-run routes within each cluster.
- Supported "brown-field" optimization, fixing selected existing facilities while optimizing the rest of the network; integrated HERE Maps routing and interactive Leaflet visualizations for operational stakeholders.

Demand Forecasting & Inventory Planning — Western Union, Grinta (Retail / Supply Planning)

- Western Union: built a branch-level time-series forecasting model for daily cash-withdrawal amounts, cutting cash over-supply error **from 300% to 20%**.
- Grinta: built a time-series forecasting model for medicine inventory replenishment, reducing carrying fees on unsold stock while avoiding missed sales from stock-outs simultaneously.

Customer Analytics — Lead Scoring & Churn Prediction — Wayak Health, AMAN (Retail / CRM)

- Wayak Health: built an end-to-end lead-scoring pipeline (feature engineering, model training, weekly batch scoring) that lifted patient-conversion targeting precision **from 3% to 25%**.
- AMAN Financial Services: wrangled **100GB+** of point-of-sale transaction data across **100K+ merchants** and developed an XGBoost churn model reaching **79% precision**, with findings delivered to stakeholders via dashboards and reports.

Pricing & Marketplace Analytics — London Cab, Naqla, Sa3ar (Retail / Dynamic Pricing)

- London Cab: built temporal clustering models to identify optimal driver waiting locations by time of day, cutting passenger pickup times **by 13%**; designed a simulation dashboard for testing pricing strategies before deployment.
- Naqla: built an analytical dashboard plus deep-learning and XGBoost regression models to estimate truck-trip prices from shipment characteristics.
- Sa3ar: scraped a market dataset from classified listings and built a fair-price estimation model for used cars from specs, mileage and condition.

Credit Risk ML Platform — 12 Financial Institutions

- Architected and deployed credit-scoring pipelines for **12 financial institutions**, processing **100K+ loan applications/month** and facilitating **$150M+ in lending**, cutting average default rates **40%**.
- Owned the complete ML lifecycle — data wrangling, EDA, feature engineering, model training, Dockerized APIs, and production support. Additional engagements: Tamweely, B.TECH, EGBank, Zeyada, Sarwa.

Generative AI & LLM Systems — Raya & Forsa

- Raya: built a multilingual GPT-4o NLP pipeline extracting income, spending and vendor-category signals from SMS for thin-file alternative credit scoring; fine-tuned Llama 3.1 8B with QLoRA to cut inference cost while holding output quality.
- Forsa: built GPT-4o-powered WhatsApp assistants (via Twilio) for activation and collections conversations, using a lightweight prompt-grounded RAG architecture to keep responses policy-compliant.

Scientific Advisor *Aug 2018 – Mar 2019*

Interstellar Consultation Services — Cairo, Egypt

- Supervised development of **dbparser**, an open-source R package for parsing and integrating public drug databases (DrugBank, KEGG) — **50K+ downloads** and active community adoption.
- Defined research directions and feature priorities aligned with drug-discovery workflows.

Software Engineer (Web Developer) *Oct 2008 – May 2011*

United OFOQ — Cairo, Egypt

- Delivered updates, fixes and on-site deployment (Amman, Jordan) for the Baggage Reconciliation System (BRS), and built the reporting functionality (JasperReports) for the OFOQ Talent Management System (OTMS).
- *Technologies: Java EE, JavaScript, IBM DB2, Oracle Database*

EDUCATION

PhD in Computer Science *Aug 2013 – Jun 2018*

Nanyang Technological University — Singapore

- Research in ML-based prediction of drug-target interactions using ensemble learning, matrix factorization, manifold learning and class-imbalance-aware methods.
- Published **4 peer-reviewed papers** and **1 book chapter**; **Best Paper Award, InCoB 2016** (BMC Bioinformatics).
- **Thesis:** *Challenges and Solutions in Drug-Target Interaction Prediction*. Graduated with a CGPA of **4.88 / 5.00**.
- **Relevant graduate coursework:** Advanced Topic in Convex Optimisation, Data Mining, Computational Intelligence: Methods & Applications, High Performance Computing.

MSc in Bioinformatics *Aug 2011 – Dec 2012*

Nanyang Technological University — Singapore

- Graduated with a CGPA of **4.69 / 5.00**. Dissertation on structure-based prediction of ligand binding site residues.
- **Relevant coursework:** Biostatistics, Algorithms for Bioinformatics, High Performance Computing, Computational Biology.

BSc in Computer and Information Sciences *Sep 2002 – Aug 2006*

Ain Shams University — Cairo, Egypt

SKILLS

- **Forecasting & Optimization:** Time-series forecasting, demand planning, inventory optimization, network/facility-location optimization, vehicle routing (TSP), clustering, recommender systems, dynamic pricing
- **Machine Learning & Deep Learning:** End-to-end ML ownership, feature engineering, XGBoost, deep learning (PyTorch), causal inference, A/B testing, association analysis
- **Generative AI & NLP:** Large Language Models, RAG, conversational AI (GPT-4o, LLaMA 3.1), prompt engineering, QLoRA fine-tuning (Unsloth)
- **Languages & Libraries:** Python, R, MATLAB, SQL · PyTorch, scikit-learn, pandas, NumPy, Transformers, LangChain, Plotly, XGBoost, tidyverse, ggplot2, Shiny
- **Infrastructure & DevOps:** AWS, Docker, Git, MLflow, Spark, Jupyter, Neo4j

SELECTED PUBLICATIONS

- Computational Prediction of Drug–Target Interactions using Chemogenomic Approaches: An Empirical Survey — Briefings in Bioinformatics, 2018
- Drug–Target Interaction Prediction with Graph Regularized Matrix Factorization — IEEE/ACM TCBB, 2017
- Drug–Target Interaction Prediction using Ensemble Learning and Dimensionality Reduction — Methods (Elsevier), 2017
- Drug–Target Interaction Prediction via Class Imbalance-Aware Ensemble Learning — BMC Bioinformatics, 2016 (Best Paper Award, InCoB 2016)
- Computational Prediction of Drug–Target Interactions via Ensemble Learning — book chapter, In Silico Methods for Drug Repurposing, Springer, 2019

CERTIFICATIONS

- **Forecasting & Optimization:** Time Series Analysis in Python · Manipulating Time Series Data in Python · Visualizing Time Series Data in Python · Introduction to Optimization in Python (DataCamp)
- **Deep Learning & NLP:** Deep Learning in Python skill track — PyTorch for images, text and Transformer models (DataCamp) · Machine Learning: Natural Language Processing (Udemy)
- **Generative AI:** Multimodal RAG — AI Search & Recommender Systems with GPT-4 · Neo4j: Cypher, GDS, GraphQL, LLM & Knowledge Graphs for RAG (Udemy)
- **Algorithms:** AlgoExpert — 100 data-structures and algorithms problems completed